

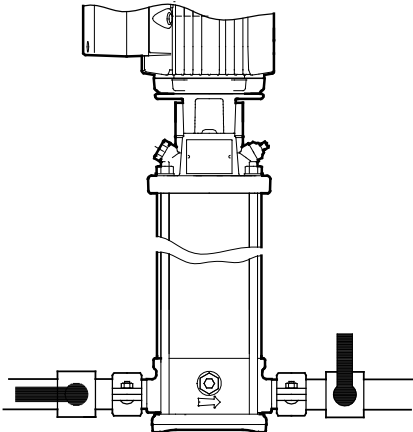
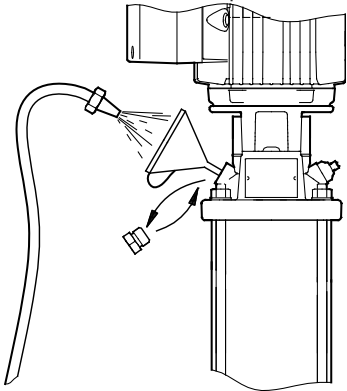
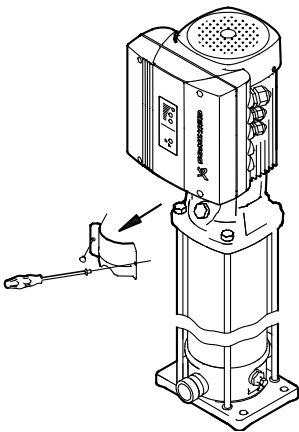
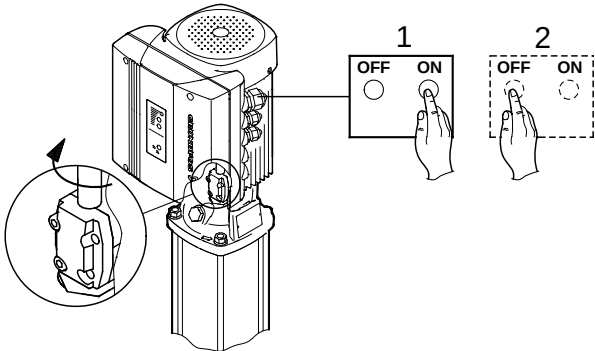
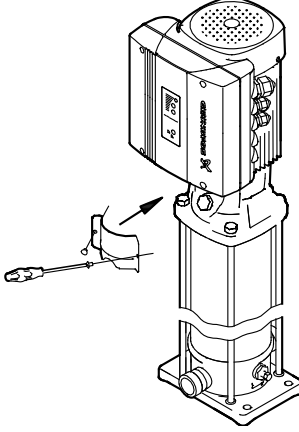
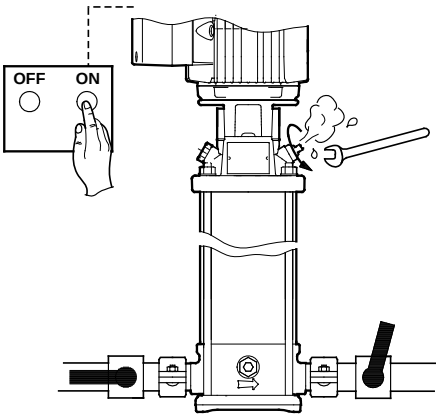
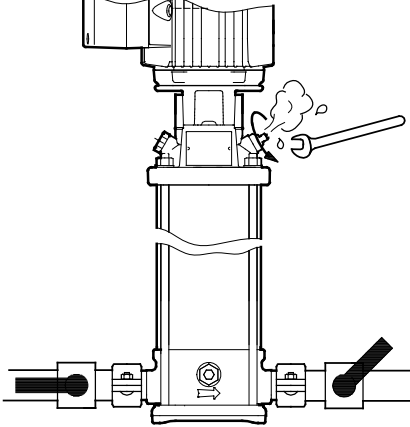
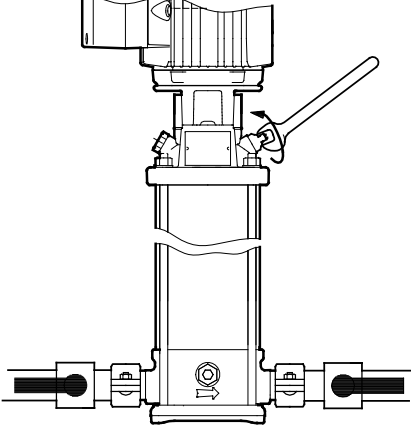
CRNE-HS

Installation and operating instructions

GB D DK RU



Start-up

1  TM03 0489 5204	2  TM03 0490 5204
3  TM03 0491 5204	4  TM03 0492 5204
5  TM03 0493 5204	6  TM03 0494 5204
7  TM03 0495 5204	8  TM03 0496 5204

GB Start-up

1 Close the isolating valve on the discharge side of the pump and open the isolating valve on the suction side.	2 Remove the priming plug from the pump head and slowly fill the pump with liquid. Replace the priming plug and tighten securely.
3 Remove a coupling guard.	4 Start the pump for a short period and check the direction of rotation. The correct direction of rotation is clockwise when seen from the non-drive end . Stop the pump. Note: The direction of rotation has been factory-set and cannot be changed.
5 Replace the coupling guard.	6 Start and vent the pump by means of the vent valve in the pump head. At the same time, open the discharge isolating valve a little.
7 Continue to vent the pump. At the same time, open the discharge isolating valve a little more.	8 Close the vent valve when a steady stream of liquid runs out of it. Completely open the discharge isolating valve.
For further information, see section 8. <i>Start-up</i> on page 8.	

D Inbetriebnahme

1 Das druckseitige Absperrventil schließen und das saugseitige Absperrventil öffnen.	2 Einfüllstopfen entfernen und Pumpe langsam auffüllen. Einfüllstopfen wieder einschrauben und fest anziehen.
3 Kupplungsschutzschirm entfernen.	4 Pumpe kurz einschalten und Drehrichtung der Pumpe prüfen. Vom Lüfter gesehen muss die Pumpe im Uhrzeigersinn laufen. Pumpe ausschalten. Hinweis: Die Drehrichtung ist werkseitig eingestellt und lässt sich nicht ändern.
5 Kupplungsschutzschirm wieder montieren.	6 Pumpe einschalten und über das Entlüftungsventil im Kopfstück entlüften. Gleichzeitig das druckseitige Absperrventil ein wenig öffnen.
7 Die Entlüftungsvorgehensweise fortsetzen. Gleichzeitig das druckseitige Absperrventil ein bisschen mehr öffnen.	8 Entlüftungsventil schließen, wenn das Medium aus dem Ventil herausläuft. Das druckseitige Absperrventil ganz öffnen.
Für weitere Informationen, siehe Abschnitt 8. <i>Inbetriebnahme</i> auf Seite 12.	

DK Idriftsætning

1 Luk afspærringsventilen på pumpens afgangsside og åbn afspærringsventilen på pumpens tilgangsside.	2 Afmontér spædeproppen i topstykket og spæd pumpen langsomt. Montér derefter spædeproppen igen.
3 Fjern en koblingsbeskyttelsesskærm.	4 Start pumpen kortvarigt og kontrollér pumpens omdrejningsretning. Set fra ventilatorsiden er korrekt omdrejningsretning med uret . Stop pumpen. Bemærk: Omdrejningsretningen er indstillet fra fabrikken og kan ikke ændres.
5 Montér koblingsbeskyttelsesskærmen.	6 Start og udluft pumpen ved hjælp af udluftningsventilen, som er placeret i topstykket. Åbn samtidig afspærringsventilen på pumpens afgangsside lidt.
7 Fortsæt med at udlufte pumpen. Åbn samtidig afspærringsventilen på pumpens afgangsside lidt mere.	8 Luk udluftningsventilen, når der løber en jævn væskestrøm ud af den. Åbn afspærringsventilen på pumpens afgangsside helt.
For yderligere information, se afsnit 8. <i>Idriftsætning</i> på side 16.	

1 Перекройте запорный клапан со стороны выпуска насоса и откройте запорный клапан со стороны всасывания.	2 Выньте пробку из заливочной горловины и медленно заполните насос жидкостью. Установите пробку заливочной горловины на свое место и прочно затяните.
3 Снимите соединительную защитную пластинку.	4 Запустите насос и дайте ему поработать в течение короткого периода времени. Проверьте направление вращения. Правильным направлением вращения является направление по часовой стрелке, если смотреть со стороны конца, который не соединен с приводом. Остановите насос. Примечание: Направление вращения устанавливается изготовителем при сборке и не может быть изменено.
5 Установите соединительную защитную пластинку на свое место.	6 Запустите и продуйте насос при помощи выпускного клапана, который находится на уровне напора насоса. Одновременно слегка приоткройте запорный клапан со стороны выпуска.
7 Продолжайте вентиляцию насоса. Одновременно слегка продолжайте приоткрывать запорный клапан со стороны выпуска.	8 Закройте выпускной клапан, когда из насоса начнет вытекать жидкость ровным потоком. Полностью откройте запорный клапан со стороны выпуска.
Что касается дальнейшей информации, обратитесь к разделу 8. <i>Ввод в эксплуатацию</i> , который находится на странице 20.	

CRNE-HS

Installation and operating instructions

6

GB

Montage- und Betriebsanleitung

10

D

Monterings- og driftsinstruktion

14

DK

Руководство по монтажу и эксплуатации

18

RU

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**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document**Warning**

If these safety instructions are not observed, it may result in personal injury

**Warning**

If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

Note

Notes or instructions that make the job easier and ensure safe operation.

2. General information

This document is a supplement to these installation and operating instructions:

- CR, CRI, CRN (96462123)
- CRE, CRIE, CRNE, SPKE, MTRE, CHIE, CME (96780071).

3. Type designation

Example	CRNE 1 - 23 - HS - P - GI - E - HUUE					
Type range						
Rated flow rate in m ³ /h n = 3000 min ⁻¹						
Number of impellers						
Code for pump version						
Code for pipework connection						
Code for materials						
Code for rubber parts						
Code for shaft seal						

4. Applications

The Grundfos CRNE-HS pump is designed to handle pressures up to 48 bar. The pump is fitted with a frequency-controlled motor with a speed of up to 5440 min⁻¹ depending on motor size. See section 14.6 *Maximum speed and sound pressure level*.

The pump is typically used as a pressure booster pump in systems with variable demands.

On delivery, the pump is ready for installation and start-up. The pump will start up at 100 % performance.

5. Operating conditions**5.1 Temperature ranges****Ambient temperature**

-20 °C to +40 °C.

Liquid temperature

-20 °C to +120 °C (HQQV shaft seal: -20 °C to +90 °C).

Caution

The liquid temperature applies to the pump only.

Temperature during storage/transport

-40 °C to +60 °C.

Relative air humidity

Maximum 95 %.

5.2 Inlet pressure

Minimum inlet pressure

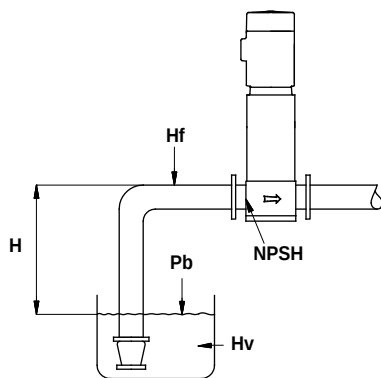


Fig. 1 Schematic view of open system with a CRNE-HS pump

The maximum suction lift "H" in metres head can be calculated as follows:

$$H = p_b \times 10.2 - \text{NPSH} - H_f - H_v - H_s$$

p_b = Barometric pressure in bar.

(Barometric pressure can be set to 1 bar.)

In closed systems, p_b indicates the system pressure in bar.

NPSH = Net Positive Suction Head in metres head (to be read from the NPSH curve on page 23 at the highest flow the pump will be delivering).

H_f = Friction loss in suction pipe in metres head at the highest flow the pump will be delivering.

H_v = Vapour pressure in metres head, see Fig. B on page 23.
 t_m = Liquid temperature.

H_s = Safety margin = minimum 0.5 metres head.

If the calculated "H" is positive, the pump can operate at a suction lift of maximum "H" metres head.

If the calculated "H" is negative, an inlet pressure of minimum "H" metres head is required. There must be a pressure equal to the calculated "H" during operation.

Example

$p_b = 1$ bar.

Pump type: CRNE 1 HS.

Flow rate: 2 m³/h.

NPSH (from page 23): 2.0 metres head.

$H_f = 3.0$ metres head.

Liquid temperature: +60 °C.

H_v (from Fig. B, page 23): 2.1 metres head.

$$H = p_b \times 10.2 - \text{NPSH} - H_f - H_v - H_s \text{ [metres head]}.$$

$$H = 1 \times 10.2 - 2.0 - 3.0 - 2.1 - 0.5 = \mathbf{2.6 \text{ metres head.}}$$

This means that the pump can operate at a suction lift of maximum 2.6 metres head.

Pressure calculated in bar: $2.6 \times 0.0981 = 0.255$ bar.

Pressure calculated in kPa: $2.6 \times 9.81 = 25.5$ kPa.

Maximum inlet pressure

- 10 bar (static pressure).
- 25 bar (pressure during operation).

Maximum operating pressure

50 bar.

Caution The actual inlet pressure + the maximum pump pressure (at no flow) must not exceed 50 bar.

Caution The pump must not run against a closed discharge valve.

5.2.1 Pressure test

The pump may be tested at a static pressure of 55 bar.

Note

During the test, the pump shaft must be locked to prevent axial movement.

To lock the pump shaft, proceed as follows:

1. Remove the motor.
2. Lift the chamber stack as far as possible and fit a bracket (fig. 2) to the pump shaft and the pump head. The bracket must be ordered separately (product number 96465853).

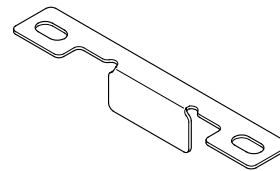


Fig. 2 Bracket for locking the pump shaft

5.3 Minimum flow rate

Pump type	Liquid temperature	
	Up to +80 °C	+80 to +90 °C
CRNE 1 HS	0.5 m ³ /h	1.0 m ³ /h
CRNE 3 HS	0.5 m ³ /h	1.0 m ³ /h

5.4 Shaft seal

The shaft seal is fitted on the suction side of the pump. As the shaft seal is not affected by the discharge pressure, the choice of shaft seal is only determined by the inlet pressure as well as the chemical properties and temperature of the liquid to be pumped.

5.5 Start/stop of pump

The number of starts and stops via the mains supply must not exceed 4 times per hour. When switched on via the mains supply, the pump will start within 5 seconds.

If a higher number of starts and stops is desired, the input for external start/stop must be used when starting/stopping the pump.

When started/stopped via an external on/off switch, the pump will start immediately.

The maximum number of starts and stops is 200 times per hour.

6. Installation

See the installation and operating instructions for the CR, CRI, CRN pumps.

Caution

Do not use rapid-closing valves, e.g. rapid-closing spray guns, as these may cause water hammering.

In order to maintain the UL/cUL approval, additional installation procedures must be followed. See page 22.

6.1 Motor cooling

To ensure cooling of motor and electronics, the following must be observed:

- Place the pump in such a way that sufficient cooling is ensured.
- Keep the temperature of the cooling air below 40 °C.
- Keep motor cooling fins and fan blades clean.

7. Electrical connection

See the installation and operating instructions for the CRE, CRIE, CRNE, SPKE, MTRE, CHIE and CME pumps.

Warning

The user or the installer is responsible for the installation of correct earthing and protection according to national and local standards. All operations must be carried out by qualified personnel.

Warning

Before making any connections in the pump terminal box, make sure that the power supply has been switched off for at least 5 minutes and that it cannot be accidentally switched on.

Note for instance that the signal relay may be connected to an external supply which is still connected when the mains supply is disconnected.

8. Start-up

Caution

Do not start the pump until it has been filled with liquid and vented. If the pump runs dry, the pump bearings and the shaft seal may be damaged.

Warning

Pay attention to the direction of the vent hole and take care to ensure that the escaping water does not cause injury to persons or damage to the motor or other components.

In hot-water installations, pay special attention to the risk of injury caused by scalding hot water.

Follow the instructions on page 2.

Note

The direction of flow in CRNE-HS pumps is the opposite of the one in the standard pump.

It is advisable to open the bypass valve during start-up. The bypass valve connects the suction and discharge sides of the pump, thus making the filling procedure easier. When the operation is stable, the bypass valve can be closed. When pumping liquids containing air, it is advisable to leave the bypass valve open.

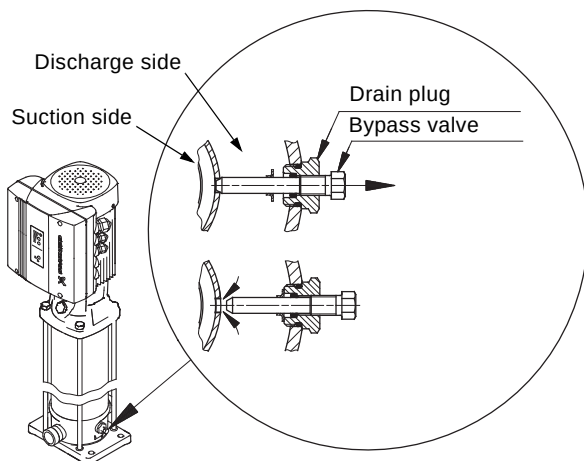


Fig. 3 Filling by means of bypass valve

TM03 0502 0105

9. Functions

The CRNE-HS pump offers the following functions:

Controlled operation

If the pump is connected to an external sensor, the following parameters can be controlled:

- pressure
- differential pressure
- temperature
- differential temperature
- flow rate.

Note *In constant-pressure mode, a pressure transmitter, if used, must be fitted after the pump – not in the pump head. This is due to the special construction of the pump.*

Uncontrolled operation

The pump performance can be set according to the demand.

Setting of the pump

The desired setpoint can be set directly on the pump control panel, via an input for external setpoint signal, via a bus or by means of the Grundfos wireless remote control R100.

All other settings are made by means of the R100.

Important parameters such as actual value of control parameter, power consumption, etc. can be read via the R100.

Inputs

The pump incorporates inputs for external potential-free contacts for start/stop and digital function. The digital function enables external setting of max. curve, min. curve, external fault function or flow switch.

Output

The pump incorporates an output for a potential-free fault, operating or ready signal.

Bus communication

The pump incorporates an input for bus communication.

Via the bus communication input, the pump can be controlled and monitored by a building management system or another external control system.

10. Setting the pump

See the installation and operating manual for the CRE, CRIE, CRNE, SPKE, MTRE, CHIE and CME pumps.

11. Maintenance

Warning

Before starting work on the pump, make sure that the power supply has been switched off for at least 5 minutes and that it cannot be accidentally switched on.

The pump bearings, the shaft seal and the motor bearings are maintenance-free.

If the pump is to be drained for a long period of inactivity, remove one of the coupling guards to inject a few drops of silicone oil on the shaft between the pump head and the coupling. This will prevent the shaft seal faces from sticking.

12. Megging

Caution

Megging of an installation incorporating CRNE-HS pumps is not allowed, as the built-in electronics may be damaged.

13. Coupling adjustment

For adjustment of coupling of CRNE 1 and 3, see Fig. D on page 25.

14. Technical data

14.1 Supply voltage

3 x 380-480 V - 10 %/+ 10 %, 50/60 Hz, PE.

Cable: 6-10 mm²/10-8 AWG.

See nameplate.

Recommended fuse sizes

Motor sizes from 4.6 to 6.0 kW: Max. 16 A.

Motor size 7.5 kW: Max. 32 A.

Standard as well as quick-blow or slow-blow fuses may be used.

14.2 Leakage current

Motor size [kW]	Leakage current [mA]
4.6 and 6.0	< 5
7.5	< 10

The leakage currents are measured in accordance with EN 60355-1.

14.3 Enclosure class

Standard: IP55 (IEC 34-5).

14.4 Insulation class

F (IEC 85).

14.5 Dimensions and weights

Dimensions: See *Fig. C* and page 24.

Weights: See label on the packing.

14.6 Maximum speed and sound pressure level

Motor size [kW]	CRNE-HS 1-23		CRNE-HS 3-23	
	Speed [min ⁻¹]	Sound pressure [dBA]	Speed [min ⁻¹]	Sound pressure [dBA]
4.6	4800	72	4050	67
6.0	5180	74	4450	70
7.5	5440	75	4750	71

The sound pressure level is calculated according to ISO 9871 and ISO/DIS 11203, method Q2.

15. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

1. Installation in the USA and Canada

In order to maintain the UL/cUL approval, these additional installation procedures must be followed. The UL approval is according to UL508C.

1.1 Electrical installation

1.1.1 Conductors

Use 140/167 °F (60/75 °C) copper conductors only.

1.1.2 Torques

Power terminal, M4: 2.35 Nm.

Relay, M2.5: 0.5 Nm.

Input control, M2: 0.2 Nm.

1.1.3 Line reactors

Max. line reactor size, 2 unit %.

1.1.4 Fuse size/circuit breaker

The pump is "suitable for use on a circuit capable of delivering not more than 5000 A RMS symmetrical amperes, 480 V maximum"

- when protected by fuses rated (see tables below) and 480 V, and/or
- "when protected by a circuit breaker having an interrupting rating not less than (see table below) RMS symmetrical amperes, 480 V maximum".

USA - hp

2-pole	Fuse size	Circuit breaker type/model
6.2	40 A	40 A / Inverse time
8.0	40 A	40 A / Inverse time
10	50 A	50 A / Inverse time

Europe - kW

2-pole	Fuse size	Circuit breaker type/model
4.6	40 A	40 A / Inverse time
6.0	40 A	40 A / Inverse time
7.5	50 A	50 A / Inverse time

1.1.5 Overload protection

Degree of overload protection provided internally by the drive, in percent of full-load current: 102 %.

1.2 Before starting the pump

- The pump must always be connected to the power supply at least one hour prior to start.
- From then on the pump must always be connected to the power supply.

This procedure will help to avoid condensation in the terminal box.

Fig. A

NPSH

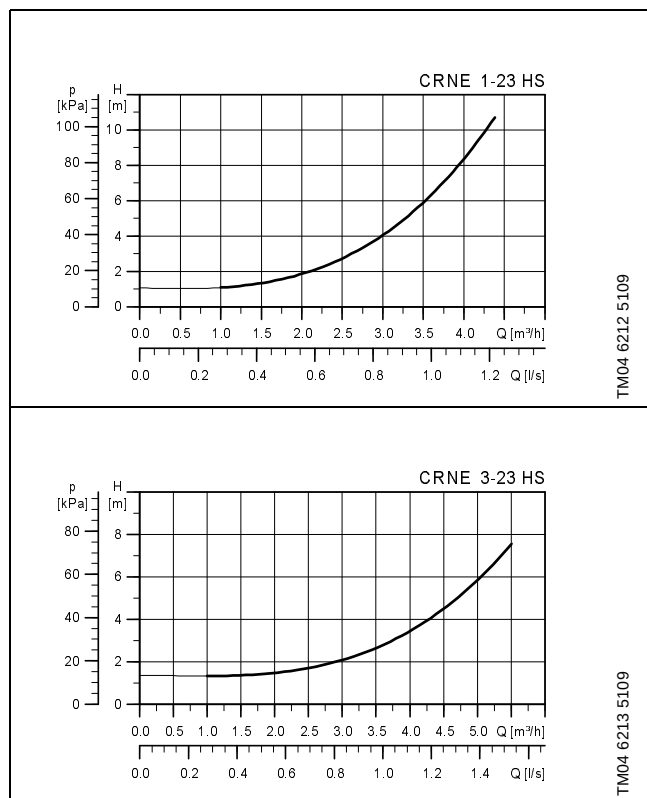
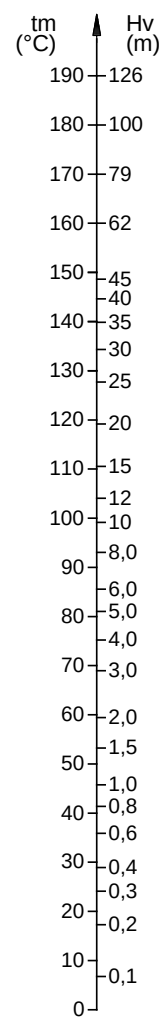


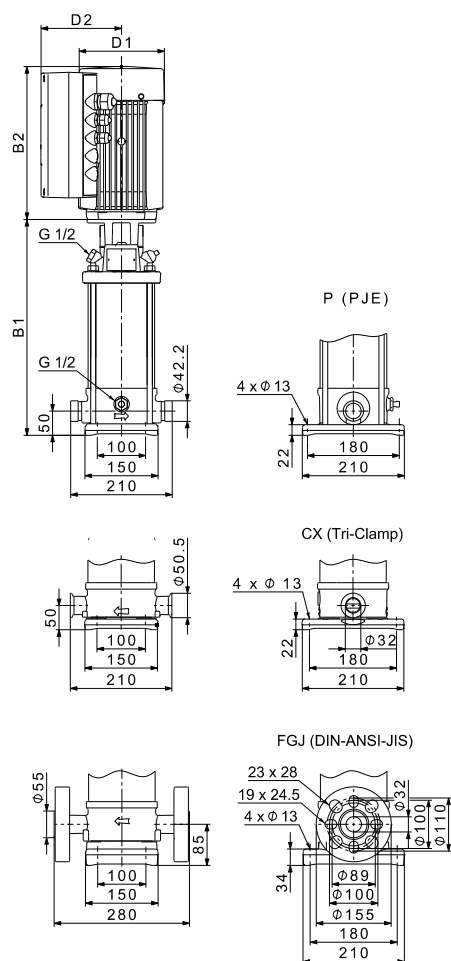
Fig. B



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Fig. C

Dimensional sketch

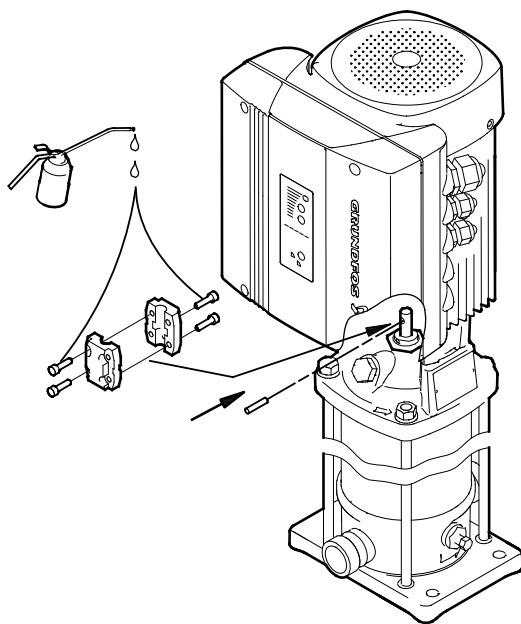


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Dimensions and weights

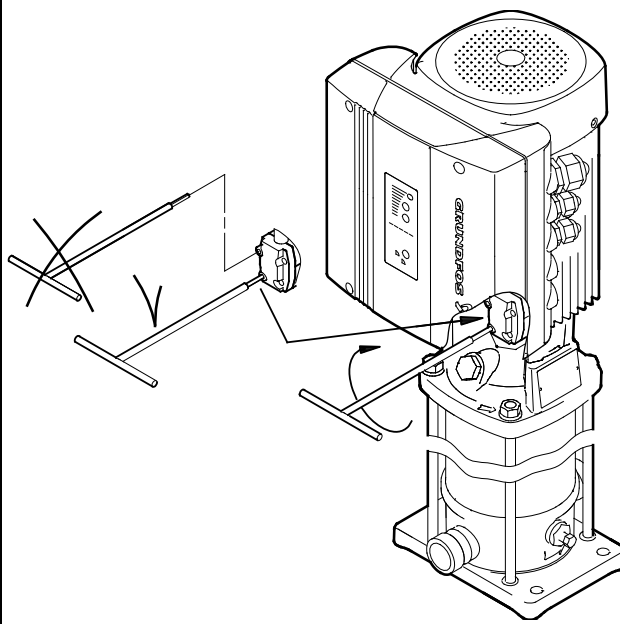
Pump type	Motor [kW]	Dimensions [mm]						Net weight [kg]	
		PJE/CX		FGJ		D1	D2	PJE/CX	PJE
		B1	B1 + B2	B1	B1 + B2				
CRNE 1-23 HS	4.6	680	1052	715	1087	220	188	62	69
CRNE 1-23 HS	6.0	680	1071	715	1106	220	188	66	73
CRNE 1-23 HS	7.5	680	1071	715	1106	260	213	71	78
CRNE 3-23 HS	4.6	680	1052	715	1087	220	188	62	69
CRNE 3-23 HS	6.0	680	1071	715	1106	220	188	62	69
CRNE 3-23 HS	7.5	680	1071	715	1106	260	213	71	78

1



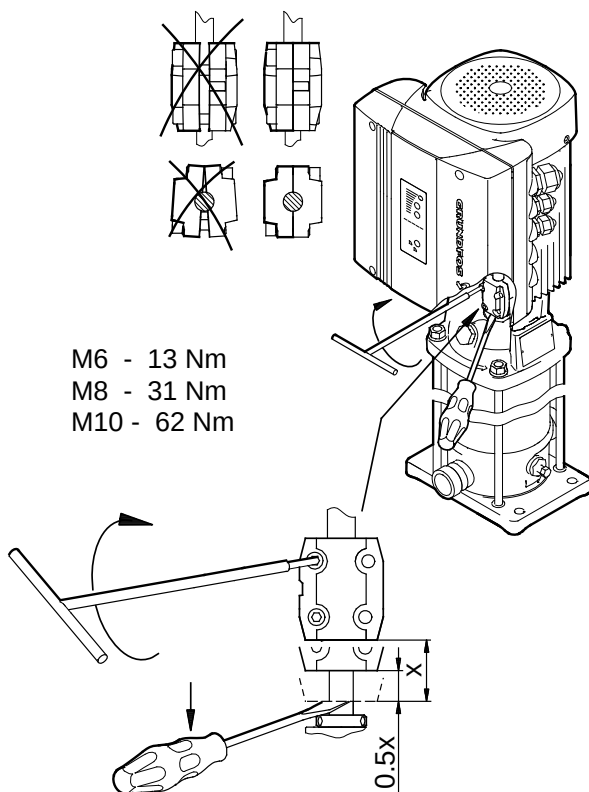
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2



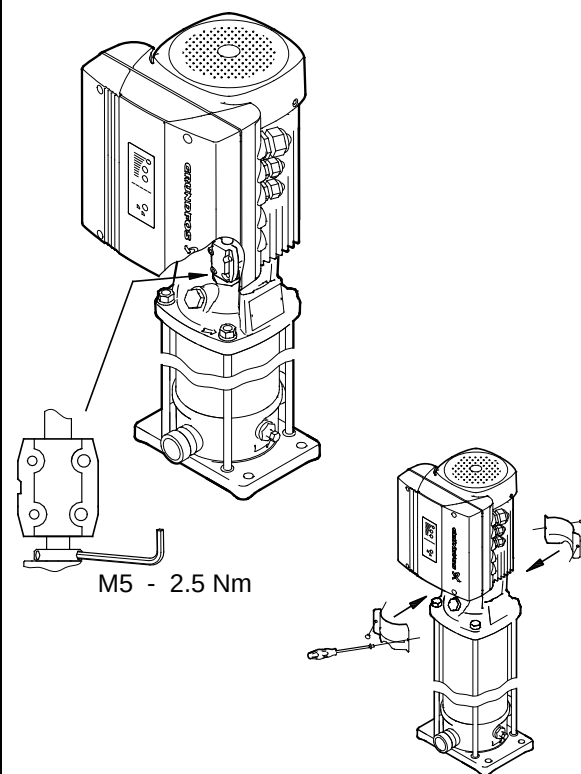
TM03 0498 5204

3



TM03 0499 5204

4



TM03 0500 5204

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